STM32F407IGT6_A

U1A

PA0-WKUP<>USART2_CTS/UART4_TX/ETH_MII_CRS/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR<>ADC123_IN0/WKUP
PA1<>USART2_RTS/UART4_RX/ETH_RMII_REF_CLK/ETH_MII_RX_CLK/TIM5_CH2/TIM2_CH2/ADC123_IN1
PA2<>USART2_TX/TIM5_CH3/TIM9_CH1/TIM2_CH3/ETH_MDIO/ADC123_IN2
PA3<>USART2_RX/TIM5_CH4/TIM9_CH2/TIM2_CH4/OTG_HS_ULPI_D0/ETH_MII_COL/ADC123_IN3
PA4<>SPI1_NSS/SPI3_NSS/USART2_CK/DCMI_HSYNC/OTG_HS_SOF/I2S3_WS<>ADC12_IN4/DAC1_OUT
PA5<>SPI1_SCK/OTG_HS_ULPI_CK/TIM2_CH1_ETR/TIM8_CHIN<>ADC12_IN5/DAC2_OUT
PA6<>SPI1_MISO/TIM8_BKIN/TIM13_CH1/DCMI_PIXCLK/TIM3_CH1/TIM1_BKIN/ADC12_IN6
PA7<>SPI1_MOSI/TIM8_CH1N/TIM14_CH1/TIM3_CH2/ETH_MII_RX_DV/TIM1_CH1N/RMII_CRS_DV/ADC12_IN7

PA8<>MCO1/USART1_CK/TIM1_CH1/I2C3_SCL/OTG_FS_SOF
PA9<>USART1_TX/TIM1_CH2/I2C3_SMBA/DCMI_D0<>OTG_FS_ VBUS
PA10<>USART1_RX/TIM1_CH3/OTG_FS_ID/DCMI_D1
PA11<>USART1_CTS/CAN1_RX/TIM1_CH4/OTG_FS_DM
PA12<>USART1_RTS/CAN1_TX/TIM1_ETR/OTG_FS_DP
PA13<>JTMS-SWDIO
PA14<>JTCK-SWCLK
PA15<>JTDI/SPI3_NSS/I2S3_WS/TIM2_CH1_ETR/SPI1_NSS

PB0<>TIM3_CH3/TIM8_CH2N/OTG_HS_ULPI_D1/ETH_MII_RXD2/TIM1_CH2N/ADC12_IN8
PB1<>TIM3_CH4/TIM8_CH3N/OTG_HS_ULPI_D2/ETH_MII_RXD3/OTG_HS_INTN/TIM1_CH3N/ADC12_IN9
PB2-BOOT1
PB3<>JTDO/ TRACESWO/SPI3_SCK/I2S3_SCK/TIM2_CH2/SPI1_SCK
PB4<>NJTRST/SPI3_MISO/TIM3_CH1/SPI1_MISO
PB5<>I2C1_SMBA/CAN2_RX/OTG_HS_ULPI_D7/ETH_PPS_OUT/TIM3_CH2/SPI1_MOSI/SPI3_MOSI/DCMI_D10/I2S3_SD
PB6<>I2C1_SCL/TIM4_CH1/CAN2_TX/OTG_FS_INTN/DCMI_D5/USART1_TX
PB7<>I2C1_SDA/FSMC_NL/DCMI_VSYNC/USART1_RX/TIM4_CH2

PB8<>TIM4_CH3/SDIO_D4/TIM10_CH1/DCMI_D6/OTG_FS_SCL/ETH_MII_TXD3/I2C1_SCL/CAN1_RX
PB9<>SPI2_NSS/I2S2_WS/TIM4_CH4/TIM11_CH1/OTG_FS_SDA/SDIO_D5/DCMI_D7/I2C1_SDA/CAN1_TX
PB10<>SPI2_SCK/I2S2_SCK/I2C2_SCL/USART3_TX/OTG_HS_ULPI_D3/ETH_MII_RX_ER/OTG_HS_SCL/TIM2_CH3
PB11<>I2C2_SDA/USART3_RX/OTG_HS_ULPI_D4/ETH_RMII_TX_EN/ETH_MII_TX_EN/OTG_HS_SDA/TIM2_CH4
PB12<>SPI2_NSS/I2S2_WS/I2C2_SMBA/USART3_CK/TIM1_BKIN/CAN2_RX/OTG_HS_ULPI_D5/ETH_RMII_TXD0/ETH_MII_TXD0/OTG_HS_ID
PB13<>SPI2_SCK/I2S2_SCK/USART3_CTS/TIM1_CH1N/CAN2_TX/OTG_HS_ULPI_D6/ETH_RMII_TXD1/ETH_MII_TXD1<>OTG_HS_ VBUS
PB14<>SPI2_MISO/TIM1_CH2N/TIM12_CH1/OTG_HS_DM/USART3_RTS/TIM8_CH2N
PB15<>SPI2_MOSI/I2S2_SD/TIM1_CH3N/TIM8_CH3N/TIM12_CH2/OTG_HS_DP/RTC_50Hz

PC0<>OTG_HS_ULPI_STP/ADC123_ IN10
PC1<>ETH_MDC/ADC123_ IN11
PC2<>SPI2_MISO/OTG_HS_ULPI_DIR/ETH_MII_TXD2/ADC123_ IN12
PC3<>SPI2_MOSI/I2S2_SD/OTG_HS_ULPI_NXT/ETH_MII_TX_CLK/ADC123_ IN13
PC4<>ETH_RMII_RX_D0/ETH_MII_RX_D0/ADC12_IN14
PC5<>ETH_RMII_RX_D1/ETH_MII_RX_D1/ADC12_IN15
PC6<>SPI2_MCK/TIM8_CH1/SDIO_D6/USART6_TX/DCMI_D0/TIM3_CH1
PC7<>SPI3_MCK/TIM8_CH2/SDIO_D7/USART6_RX/DCMI_D1/TIM3_CH2

PC8<>TIM8_CH3/SDIO_D0/TIM3_CH3/USART6_CK/DCMI_D2
PC9<>I2S2_CKIN/I2S3_CKIN/MCO2/TIM8_CH4/SDIO_D1/I2C3_SDA/DCMI_D3/TIM3_CH4
PC10<>SPI3_SCK/I2S3_SCK/UART4_TX/SDIO_D2/DCMI_D8/USART3_TX
PC11<>UART4_RX/ SPI3_MISO/SDIO_D3/DCMI_D4/USART3_RX
PC12<>UART5_TX/SDIO_CK/DCMI_D9/SPI3_MOSI/I2S3_SD/USART3_CK
PC13<>RTC_AF1
PC14<>OSC32_IN
PC15<>OSC32_OUT

NC/PDR_ON

VCAP_1

VCAP_2

VBAT
VREF+
VDDA
VSSA

PH0<>OSC_IN
PH1<>OSC_OUT
PH2<>ETH_MI_CRS
PH3<>ETH_MI_COL
PH4<>I2C2_SCL/OTG_HS_ULPI_NXT
PH5<>I2C2_SDA
PH6<>I2C2_SMBA/TIM12_CH1/ETH _MII_RXD2
PH7<>I2C3_SCL /ETH _MII_RXD3
PH8<>I2C3_SDA /DCMI_HSYNC
PH9<>I2C3_SMBA/TIM12_CH2/DCMI_D0
PH10<>TIM5_CH1TIM5_ETR/DCMI_D1
PH11<>TIM5_CH2 /DCMI_D2
PH12<>TIM5_CH3 /DCMI_D3
PH13<>TIM8_CH1N /CAN1_TX/
PH14<>TIM8_CH2N/DCMI_D4
PH15<>TIM8_CH3N/DCMI_D11

PI0<>TIM5_CH4 /SPI2_NSS/ I2S2_WS/DCMI_D13
PI1<>SPI2_SCK/I2S2_CK/DCMI_D8
PI2<>TIM8_CH4/SPI2_MISO/I2S2ext_SD/DCMI_D9
PI3<>TIM8_ETR/SPI2_MOSI/I2S2_SD/DCMI_D10
PI4<>TIM8_BKIN /DCMI_D5
PI5<>TIM8_CH1/DCMI_VSYNC
PI6<>TIM8_CH2 /DCMI_D6
PI7<>TIM8_CH3 /DCMI_D7
PI8
PI9<>CAN1_RX
PI10<>ETH _MII_RX_ER
PI11<>OTG_HS_ULPI_DIR

BOOT0<>VPP
NRST

VDD_1 VDD_2 VDD_3 VDD_4 VDD_5 VDD_6 VDD_7 VDD_8 VDD_9 VDD_10 VDD_11 VDD_12 VDD_13 VDD_14 VDD_15 → 3V3_MCU

VSS_1 VSS_2 VSS_3 VSS_4 VSS_5 VSS_6 VSS_7 VSS_8 VSS_9 VSS_10 VSS_11 VSS_12 → GND

STM32F407IGT6

RESET

S1 SW-PB: GND, NRESET
R4 10K: 3V3_MCU
C4 100NF: NRESET, GND

KEY

S2 SW-PB: 3V3_MCU, PA0
S3 SW-PB: 3V3_MCU, BOOT0

REFERENCE VOLTAGE

P1 Header 3: 3V3_MCU, VREF
R5 10R
C5 100NF, C6 10UF: VREF+, GND
R6 10R
C7 100NF, C8 10UF: VDDA, GND

START MODE

默认不焊接
NC/PDR_ON — R7 0R, R8 0R — GND, 3V3_MCU
BOOT0
BOOT1 — R9 0R, R10 0R — GND

BOOT0和BOOT1默认都用电阻连接到GND,从用户代码启动BOOT0同时和KEY0连接，按下KEY0，可以把BOOT0拉高！！方便进BootLoader模式！

FILTER CAP

3V3_MCU: C13 100NF, C14 100NF, C15 100NF, C16 100NF, C17 100NF → GND
3V3_MCU: C20 100NF, C21 100NF, C22 100NF, C23 100NF, C24 100NF → GND
3V3_MCU: C25 100NF, C26 100NF, C27 100NF, C28 100NF, C29 100NF → GND

USB OTG

USB1OTG_N R12 10R
USB1OTG_P R13 10R
GND, 5V_USB, SHELL, VBUS, SBU2, CC1, DN2, DP1, DN1, DP2, SBU1, CC2, VBUS, GND, SHELL

PULSE CIRCUIT

OSC_IN — C12 20PF — GND
Y1 8/25MHz
R11 10M
OSC_OUT — C19 20PF — GND

OSC32_IN — C11 6PF — GND
Y2 32.768KHz
OSC32_OUT — C18 6PF — GND

SWD MODE

P2 Header 4: 3V3_MCU, SW_DIO, SW_CLK, GND

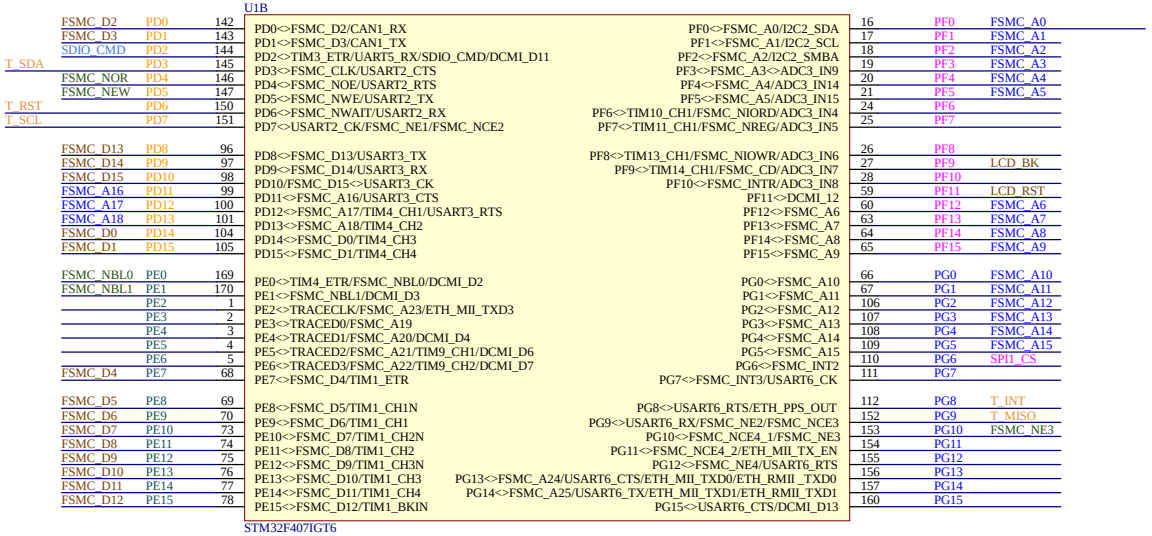
SYSTEM POWER

5V_USB — F1 1A — VCC_5V
U2 1117-3.3V: VIN, VOUT, GND → 3V3_MCU
C30 100NF, C31 10UF, C32 100NF, C33 10UF

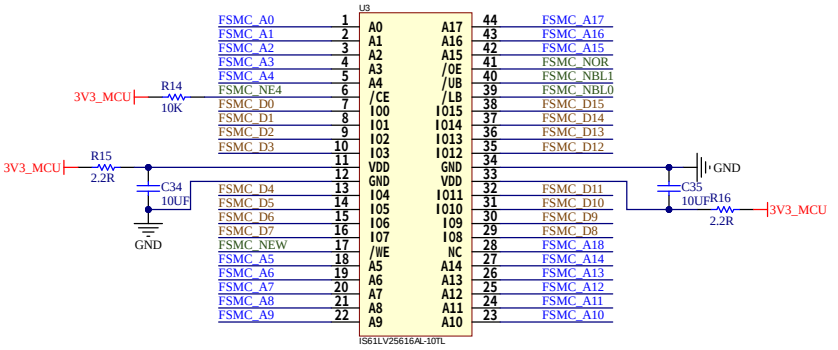
RTC POWER

VBAT — BT1 3V — 3V3_MCU, GND

STM32F407IGT6_B



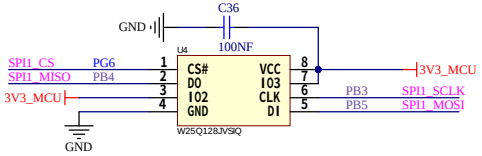
SRAM



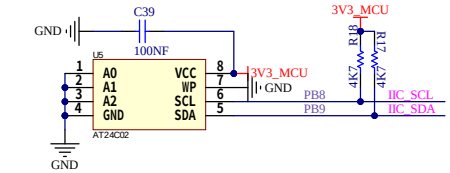
FSMC引脚名称	对应的SRAM引脚名称	说明
FSMC_NBL[1:0]	LB#、UB#	数据掩码信号
FSMC_A[18:0]	A[18:0]	行地址线
FSMC_D[15:0]	I/O[15:0]	数据线
FSMC_NWE	WE#	写入使能
FSMC_NOE	OE#	输出使能(读使能)
FSMC_NE[1:4]	CE#	片选信号

信号线	类型	说明
A0~A18	I	地址输入
I/O0~I/O7	I/O	数据输入输出信号，低字节
I/O8~I/O15	I/O	数据输入输出信号，高字节
CS2 和CS1#	I	片选信号，CS2高电平有效，CS1#低电平有效，部分芯片只有其中一个
OE#	I	输出使能信号，低电平有效
WE#	I	写入使能，低电平有效
UB#	I	数据掩码信号Upper Byte，高位字节允许访问，低电平有效
LB#	I	数据掩码信号Lower Byte，低位字节允许访问，低电平有效

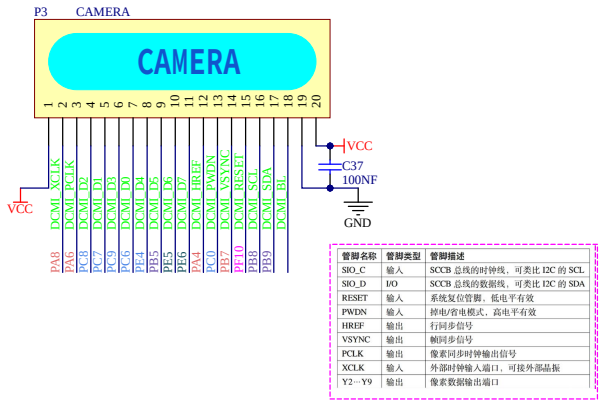
FLASH



EEPROM

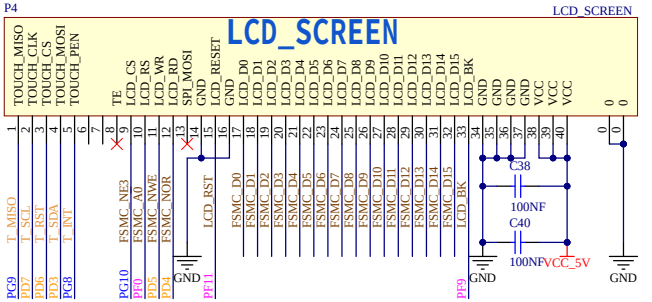


CAMERA

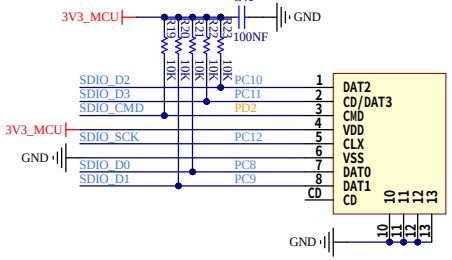


管脚名称	管脚类型	管脚描述
SIO_C	输入	SCCB 总线的时钟线，可类比 I2C 的 SCL
SIO_D	I/O	SCCB 总线的数据线，可类比 I2C 的 SDA
RESET	输入	系统复位管脚，低电平有效
PWDN	输入	掉电/省电模式，高电平有效
HREF	输出	行同步信号
VSYNC	输出	帧同步信号
PCLK	输出	像素同步时钟输出信号
XCLK	输入	外部时钟输入端口，可接外部晶振
Y2···Y9	输出	像素数据输出端口

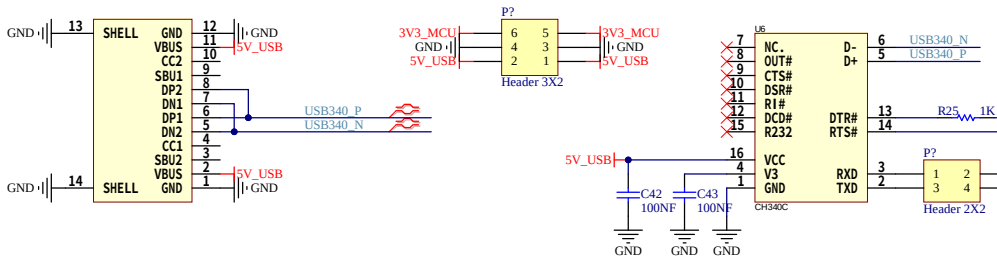
LCD SCREEN



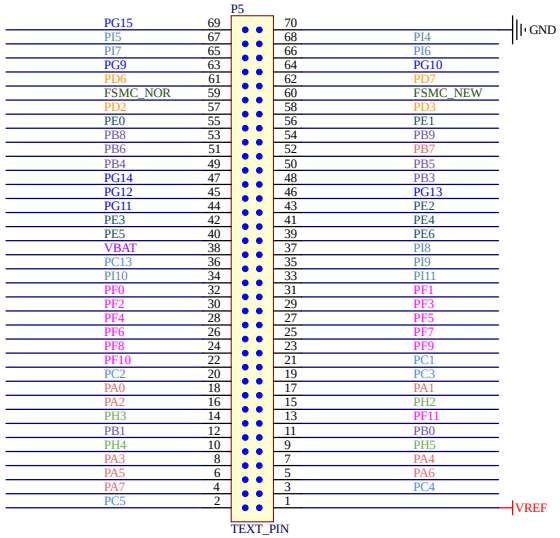
SD CARD

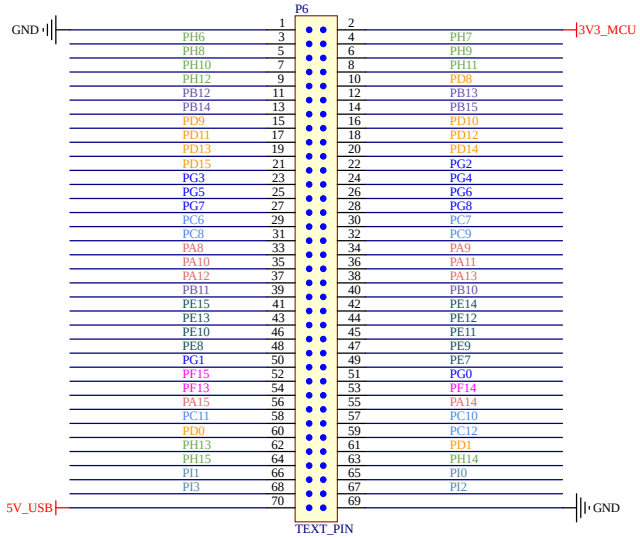


USB-TTL & DOWNLOAD

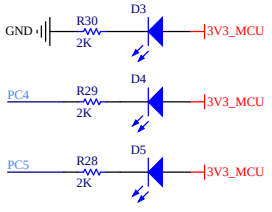


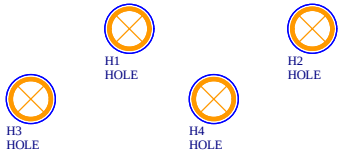
TEST_PORT

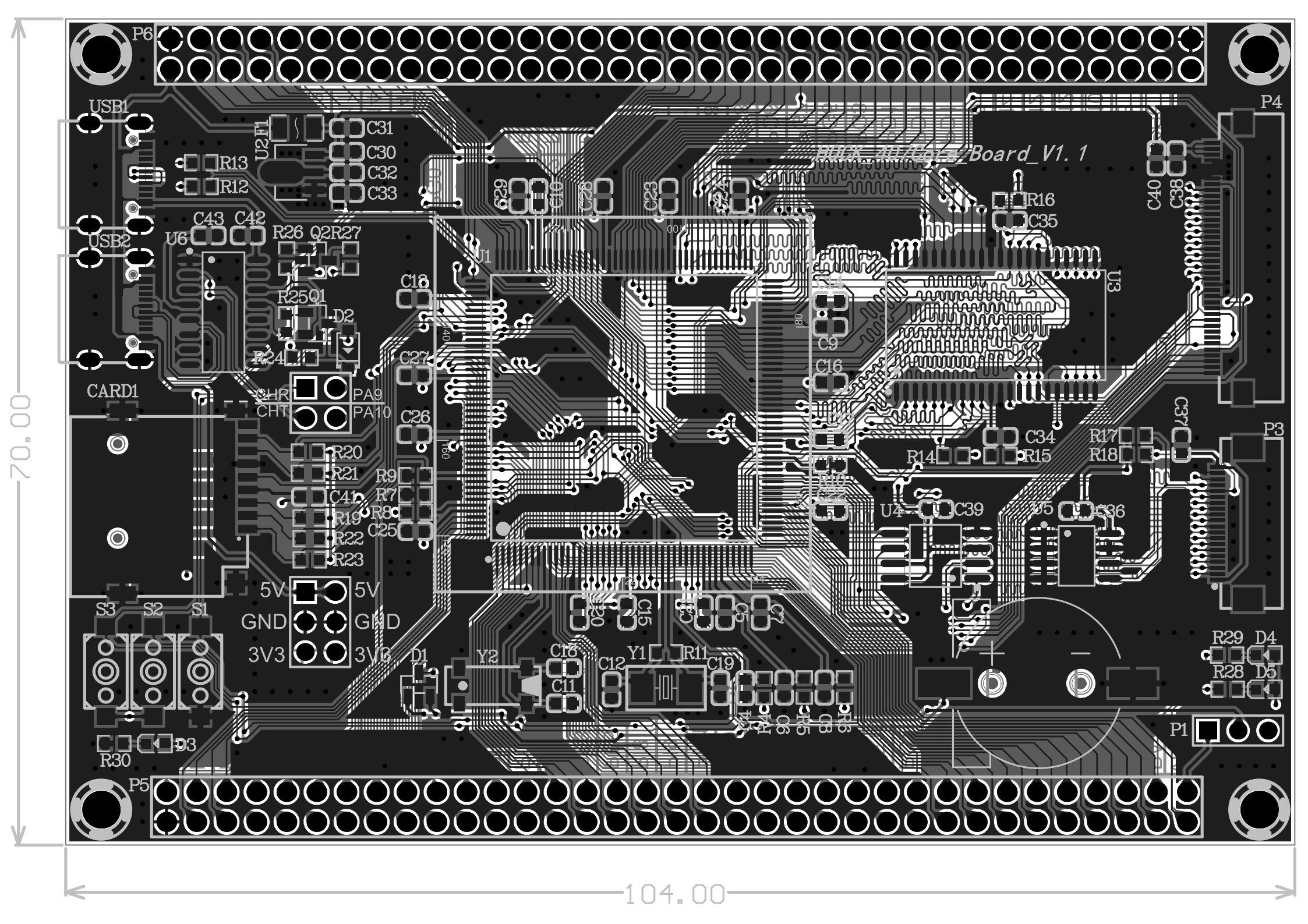




TEST LED






70.00
104.00
P6
P5
P4
P3
P1
USB1
USB2
CARD1
U1
U2
U3
U4
U5
U6
F1
Q1
Q2
D1
D2
D3
D4
D5
Y1
Y2
S1
S2
S3
R13
R12
R26
R27
R25
R24
R20
R21
R19
R22
R23
R9
R7
R8
R16
R14
R15
R17
R18
R29
R28
R30
R11
C31
C30
C32
C33
C43
C42
C16
C27
C26
C41
C25
C35
C34
C39
C36
C40
C38
C37
C15
C18
C11
C12
C19
PA9
PA10
5V
GND
3V3